

Evolution of Agency in Humanitarian Communication: A Computational Text Analysis of GlobalGiving Project Descriptions

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1 Introduction

In her seminal work, Lilie Chouliaraki (Chouliaraki, 2010) identifies three distinct phases in humanitarian communication evolution: "shock effect" appeals, "positive image" appeals, and "post-humanitarian" communication. "Shock effect" appeals rely on graphic depictions of suffering to provoke guilt, positioning beneficiaries primarily as passive victims (Benthall, 1993; Lissner, 1981). In contrast, "positive image" appeals represent beneficiaries as dignified agents, seeking to establish empathy and gratitude (Lidchi, 1999; Smillie, 1995). "Post-humanitarian" communication breaks with both approaches by emphasizing technologized action and reflexive awareness of representation itself.

Central to this evolution is the changing representation of beneficiary agency. "Shock effect" appeals position beneficiaries as passive victims, while "positive image" appeals emphasize their capacity for self-determination. Both approaches have been critiqued: the former for dehumanizing sufferers through a colonial gaze, the latter for glossing over structural injustices through narratives of individual empowerment (Dogra, 2007).

This study applies computational text analysis methods to investigate whether Chouliaraki's theoretical patterns are empirically observable in contemporary humanitarian discourse. By analyzing over 25,000 project descriptions from the GlobalGiving platform, I examine how different humanitarian sectors position beneficiaries on a spectrum from passive victims to active agents, addressing three key questions:

1. How do different humanitarian themes linguistically construct beneficiaries through patterns of agency attribution?
2. Does empirical evidence support Chouliaraki’s theoretical distinction between ”shock effect” and ”positive image” communication styles?
3. What are the underlying dimensions of linguistic variation in humanitarian discourse, and how do they relate to theoretical frameworks?

2 Data and Methods

2.1 Data Source and Thematic Categorization

This study analyzes approximately 25,000 project descriptions from GlobalGiving spanning 2010-2022. Based on qualitative review of sector codes, I grouped them into four humanitarian themes:

- **Emergency Relief** (Sector codes: M99, K30, Q71) - Projects focused on immediate disaster response
- **Empowerment & Education** (Sector codes: S30, U99, G80, N60, Q35) - Projects centered on skills development
- **Protection & Rights** (Sector codes: P30, I70, R24, R26) - Projects addressing protection issues
- **Health & Wellbeing** (Sector codes: E99, E40, F99) - Projects providing health-care services

2.2 Analytical Approach

2.2.1 Sentiment Analysis

As an initial exploration, I applied the NRC Emotion Lexicon (Mohammad and Turney, 2013), which categorizes words into emotion categories including anger, fear, joy, sadness,

trust, and positive/negative sentiment. This analysis revealed distinct emotional profiles across themes, with Emergency Relief showing higher proportions of fear and negative sentiment, while Empowerment & Education demonstrated the highest positive sentiment (see Figure 1 in Appendix).

2.2.2 Dictionary-Based Agency Analysis

To operationalize Chouliaraki’s framework, I developed custom dictionaries categorizing terms according to how they position beneficiaries:

- **”Victim” Terms:** Words emphasizing vulnerability (e.g., ”victim,” ”vulnerable,” ”poor,” ”helpless,” ”desperate,” ”displaced,” ”afflicted”)
- **”Beneficiary” Terms:** Words representing recipients neutrally (e.g., ”beneficiary,” ”recipient,” ”community,” ”family,” ”people,” ”individual,” ”resident,” ”client,” ”citizen”)
- **”Agent” Terms:** Words emphasizing capacity (e.g., ”partner,” ”leader,” ”entrepreneur,” ”participant,” ”volunteer,” ”stakeholder,” ”member,” ”driver”)

For each document, I calculated the presence of terms from each category and then examined both term frequencies and co-occurrence patterns to identify how beneficiaries are contextually positioned within different themes.

2.2.3 Co-occurrence Analysis

To understand the semantic context in which key beneficiary terms appear, I conducted a window-based co-occurrence analysis using a 5-word context window. This approach identifies which words most frequently appear near terms like ”women” and ”community” across different humanitarian themes, revealing how the meaning and framing of these terms varies by context.

2.2.4 Topic Modeling

I applied Latent Dirichlet Allocation with 8 topics to identify latent thematic structures. After extracting topic-word distributions, I calculated document-topic posteriors and averaged them within each predefined theme to understand how emergent semantic domains relate to thematic categories.

2.2.5 Vector Space Analysis

Drawing inspiration from ideological scaling approaches (Rheault and Cochrane, 2020), I employed Principal Component Analysis on the TF-IDF matrix to extract key dimensions of linguistic variation. I also applied k-means clustering ($k=6$) to identify natural groupings in this semantic space.

3 Results and Discussion

3.1 Agency Attribution Patterns

The agency attribution analysis reveals distinctive patterns across humanitarian themes. As shown in Figure 2, Emergency Relief shows the highest usage of "Beneficiary Terms" (approximately 75% of documents) and "Victim Terms" (reaching 45%), while having the lowest proportion of "Agent Terms" (below 20%). This linguistic profile aligns with Chouliaraki's "shock effect" approach, positioning beneficiaries predominantly as passive recipients.

Empowerment & Education demonstrates the highest proportion of "Agent Terms" (reaching 30%) across all themes, with terms like "entrepreneurs," "students," and "farmers" appearing prominently, constructing beneficiaries as potential agents of change. This aligns with the "positive image" communication style where beneficiaries appear as active participants.

Several counterintuitive patterns emerge in the distribution of specific terms across themes (Figure 2). Terms like "vulnerable," "victims," and "poor" appear prominently in Empowerment & Education and Protection & Rights rather than concentrating mainly

in Emergency Relief. This suggests these themes may be using vulnerability framing to justify empowerment interventions, creating a strategic blend of vulnerability and agency framings. Protection & Rights shows the highest usage of "families" and "community" terms, suggesting a more collective rather than individualistic framing approach.

The consistently lower prevalence of "Agent" terms compared to "Beneficiary" and "Victim" terms represents a key finding. Even in contexts dedicated to empowerment, the language of capacity remains secondary to the language of need. This pattern suggests Chouliaraki's hypothesized evolution toward "positive image" communication remains incomplete in contemporary discourse.

3.2 Semantic Context Through Co-occurrence Analysis

Co-occurrence analysis reveals how key terms are contextually positioned across humanitarian themes, providing deeper insight into beneficiary framing than term frequencies alone.

3.2.1 Framing of "Women"

The term "women" appears in distinctly different semantic contexts across themes:

- In **Emergency Relief**, the strongest association is with "children," positioning women alongside vulnerable groups. This exemplifies what Chouliaraki calls the "politics of pity," where women are framed primarily through victimhood.
- In **Empowerment & Education**, "women" strongly co-occurs with "business," "training," and "skills," representing an economic empowerment framing that constructs women as potential economic agents.
- In **Protection & Rights**, "women" co-occurs prominently with "violence," reflecting a rights-based framing that acknowledges gender-based violence concerns.

3.2.2 Framing of "Community"

Similarly, "community" takes on different meanings across themes:

- **Emergency Relief** co-occurrences focus on basic needs ("food," "people," "support"), positioning communities primarily as recipients of aid. This reflects Chouliaraki's "shock effect" communication style where communities are framed in terms of their vulnerability.
- **Empowerment & Education** co-occurrences emphasize development-oriented terms ("project," "development," "build"), representing communities as active agents of change.
- **Health & Wellbeing** shows balanced associations between "awareness" (suggesting community agency) and "water" (indicating service delivery).
- **Protection & Rights** demonstrates a protection-focused framing of communities, with terms related to security and vulnerability.

These co-occurrence patterns provide evidence for Chouliaraki's thesis about the evolution of humanitarian communication from "distance" (victim-oriented, shock effect) to "identity" (agency-oriented, positive image) approaches. However, they also reveal the strategic mixing of these frames within individual themes, suggesting organizations navigate complex tensions between establishing need and recognizing agency.

3.3 Latent Semantic Structures

Topic modeling revealed eight distinct semantic clusters cutting across predefined themes:

- **Topic 1 (Emergency Response):** "people," "affected," "emergency," "relief"
- **Topic 2 (Women & Girls Education):** "women," "girls," "education," "training"
- **Topic 3 (Child Protection):** "children," "support," "violence"
- **Topic 4 (Community Development):** "project," "water," "rural," "community"

- **Topic 5 (Health Systems):** "health," "program," "public"
- **Topic 6 (Medical Services):** "care," "medical," "services," "treatment"
- **Topic 7 (Community Engagement):** "community," "youth," "social," "development"
- **Topic 8 (Food Security):** "families," "food," "nutrition"

Analysis of document-topic posteriors reveals that Emergency Relief concentrates in Topics 1 (20%) and 8 (15%), reflecting its focus on immediate crisis intervention. Health & Wellbeing is dominated by Topics 5 and 6 (both at 16%), while Protection & Rights focuses on Topics 2 and 3 (16% and 15%). Empowerment & Education shows a more balanced distribution with emphasis on Topic 7 (14%).

These associations provide important context for our agency analysis findings. Emergency Relief's concentration in immediate-needs topics explains its reliance on victim terminology, while Empowerment & Education's emphasis on community engagement aligns with its higher agent terminology usage. The emergence of these patterns through unsupervised learning validates our dictionary-based findings and shows these patterns reflect genuine semantic distinctions rather than analytical artifacts.

3.4 Dimensions of Linguistic Variation

Principal Component Analysis reveals two primary dimensions structuring humanitarian discourse (Figure 4). The first principal component (PC1) represents a continuum between developmental and emergency-oriented language. On the negative end are terms like "internship," "initiatives," "transform," reflecting capacity-building and structural intervention. On the positive end are terms like "food," "care," "providing," and "relief" focused on immediate needs.

The second principal component (PC2) represents a continuum between entrepreneurial and institutional approaches. The positive end features terms like "escape," "equip," "business," "entrepreneurs," reflecting individual self-sufficiency. The negative end in-

cludes terms like "public," "crises," "campaign," "educate," suggesting institutional interventions.

This dimensional structure extends Chouliaraki's framework by showing humanitarian discourse occupies a two-dimensional space rather than a simple binary. Instead of a linear progression from "shock effect" to "positive image" approaches, we see a multidimensional landscape where themes occupy distinct positions based on temporal orientation (emergency vs. development) and intervention approach (entrepreneurial vs. institutional).

Cluster analysis shows Health & Wellbeing and Empowerment & Education forming distinct linguistic clusters (100% consistency), while other themes demonstrate more mixed patterns. These findings suggest specialized humanitarian domains develop unique communication strategies not neatly aligned with general humanitarian trends.

3.5 Temporal Evolution

Temporal analysis shows "Beneficiary Terms" consistently dominating discourse across all themes (40-80%). Health & Wellbeing demonstrates the clearest evolution, with "Beneficiary Terms" showing an upward trajectory while "Victim Terms" decline, suggesting a shift from vulnerability framing toward neutral recipient framing.

Empowerment & Education shows a recent increase in "Agent Terms" around 2020-2022, rising to nearly 50% and potentially overtaking "Beneficiary Terms," indicating an emerging emphasis on beneficiary empowerment aligning with Chouliaraki's projection.

Protection & Rights maintains the most balanced pattern, suggesting rights-based approaches may maintain consistency in acknowledging both vulnerability and agency without dramatic framing shifts over time.

3.6 Methodological Implications and Limitations

This study demonstrates the value of triangulating multiple computational methods when studying complex discursive phenomena. The dictionary-based approach effectively operationalized theoretical constructs about agency, while unsupervised methods revealed

semantic patterns difficult to detect through manual coding. Cross-validation between methods increased confidence in identified patterns, with dictionary findings independently confirmed through topic modeling and vector space analysis.

An important limitation of this study is its reliance on word occurrences rather than contextual representations. While co-occurrence analysis provides some contextual information, more sophisticated approaches like word embeddings might capture deeper semantic relationships. Word embedding techniques would allow for examining the positioning of beneficiaries in vector space, potentially revealing more nuanced patterns of agency attribution than simple word counts can provide (?). Future research could apply these techniques to examine how the semantic meaning of beneficiary terms varies across different humanitarian contexts.

Additionally, dictionary methods, while effective for identifying broad patterns, cannot capture all linguistic nuances of beneficiary representation. Terms may have different connotations depending on their context, which a dictionary approach may miss. For example, "vulnerable" might signify different degrees of agency when paired with different verbs or qualifiers. More advanced natural language processing techniques could address these limitations in future research.

4 Conclusion

This study has examined how humanitarian organizations linguistically construct beneficiaries through multiple computational approaches. The findings provide substantial empirical evidence for Chouliaraki's framework while revealing a more complex landscape than her linear evolutionary model suggests.

Different humanitarian themes employ distinctive linguistic strategies, with Emergency Relief positioning beneficiaries as passive victims and Empowerment & Education representing them more as active agents. However, rather than a linear progression from "shock effect" to "positive image" communication, we observe a multidimensional discursive space where themes occupy positions based on their temporal orientation and

intervention approach.

Most significantly, even empowerment-focused communication relies heavily on vulnerability framing, with "Beneficiary" and "Victim" terms consistently outweighing "Agent" terms. This suggests the evolution toward agency-centered representation remains incomplete and constrained by humanitarian communication's moral economy. Organizations must establish need to justify intervention while demonstrating potential impact to inspire action, creating a fundamental tension that manifests in blending vulnerability and agency framing rather than shifting entirely from one approach to another.

This research contributes to understanding how organizations linguistically construct relationships between donors and beneficiaries. It demonstrates computational text analysis methods can provide valuable insights into humanitarian discourse's moral economies, illuminating how organizations navigate ethical complexities of representing distant suffering in a media-saturated environment.

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A Figures

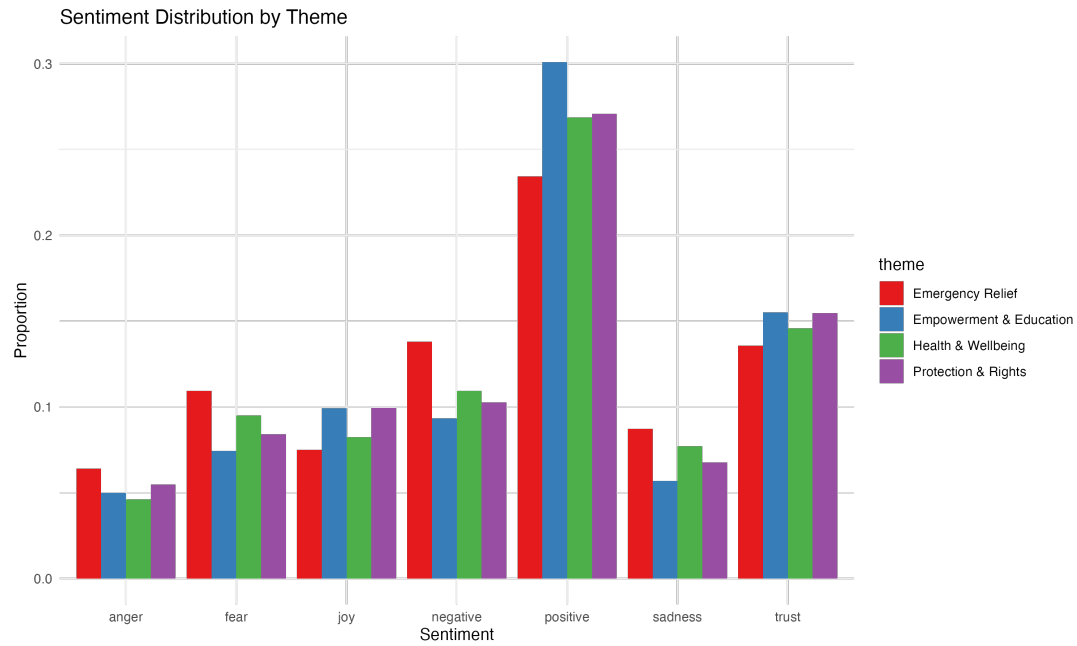


Figure 1: Sentiment Distribution by Theme. The figure shows the proportion of different sentiment categories across the four humanitarian themes. Note that Emergency Relief shows higher proportions of fear and negative sentiment, while Empowerment & Education demonstrates the highest positive sentiment.

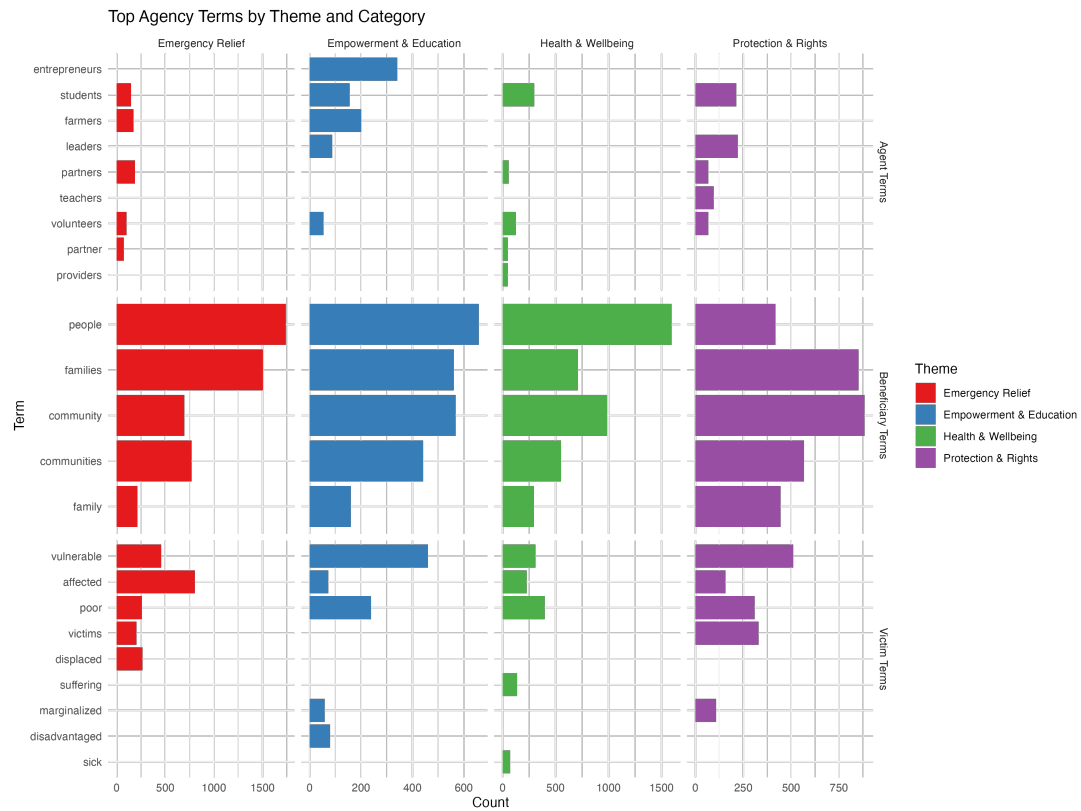


Figure 2: Top Agency Terms by Theme and Category. The figure shows the distribution of the most frequent terms in each agency category (Agent Terms, Beneficiary Terms, Victim Terms) across humanitarian themes. Note the higher proportion of "Agent" terms in Empowerment & Education, the counterintuitive appearance of "vulnerable" terminology in Empowerment narratives, and the prominence of collective terms like "families" and "community" in Protection & Rights.

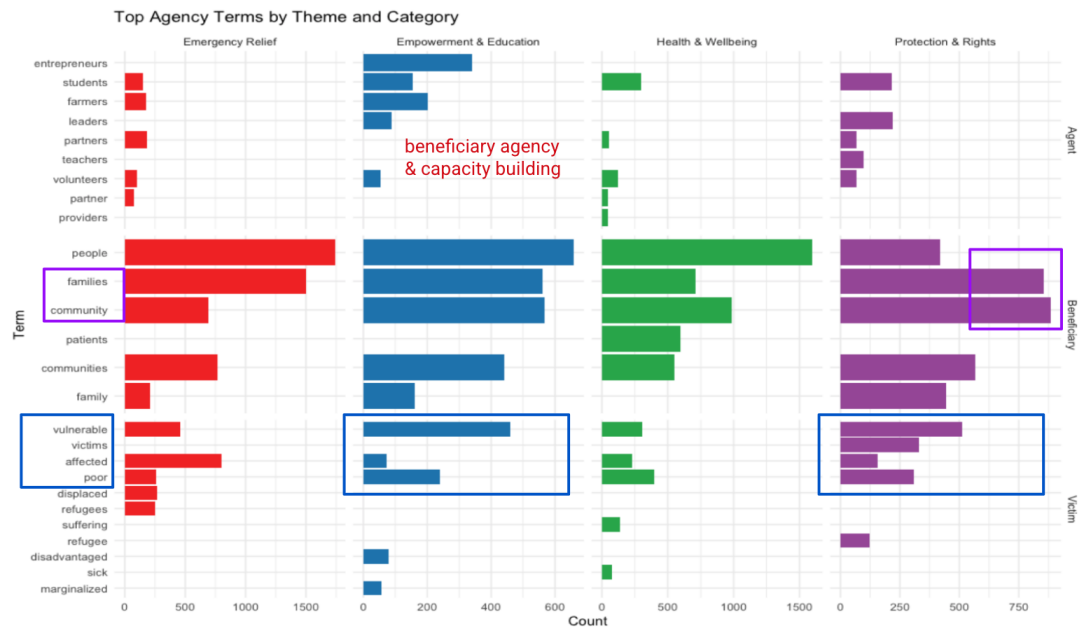


Figure 3: Annotated Agency Terms Analysis. The figure highlights key patterns in beneficiary representation across themes: collective framing in Protection & Rights (purple box), beneficiary agency and capacity building in Empowerment & Education (red annotations), and the strategic use of vulnerability terminology (blue boxes) even in empowerment-oriented contexts.

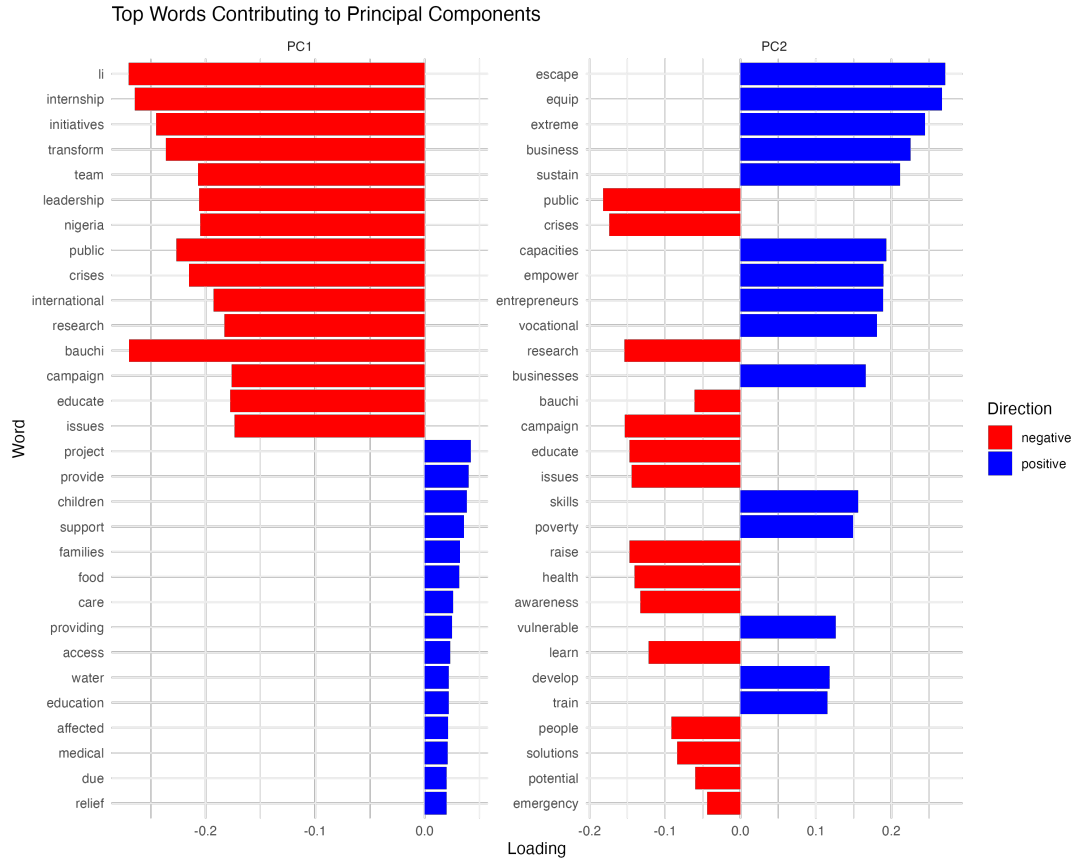


Figure 4: Top Words Contributing to Principal Components. The figure shows the words with the highest positive and negative loadings on the first two principal components derived from the TF-IDF matrix. PC1 distinguishes between developmental language (negative loadings) and emergency-oriented language (positive loadings), while PC2 separates entrepreneurial approaches (positive loadings) from institutional interventions (negative loadings).